

POE EXAM 1



JEREMIAH NYAGAH NATIONAL POLYTECHNIC

DEPARTMENT: ELECTRICAL AND ELECTRONICS ENGINEERING

EE LEVEL 6 & DCBET

SUBJECT: ELECTRICAL POWER LINES

END OF THE TERM EXAM

PERIOD: 2 HRS

Instruction:

- a) Attempt **ALL** the **QUESTIONS**
- b) Any form of cheating is not allowed
- c) Phone are not allowed in the exam room

QUESTIONS

- 1. State any five elements of a transmission line. (5 marks)
- 2. Differentiate a long transmission line, medium transmission line and short transmission line. (3 marks)
- 3. Explain a transmission Line. (2 marks)
- 4. Give four types of line supports. (4 marks)
- 5. State three materials used for making line supports. (3 marks)
- 6. State Five types of transmission line conductors. (5 marks)
- 7. Describe factors affecting the sag. (5 marks)
- 8. State four types of electrical joints. (4 marks)
- 9. State three type of electrical load. (3 marks)

10. What do you understand by the term “transmission line termination.” (2 marks)
11. Describe three types of transmission line termination. (3 marks)
12. What is the importance of termination? (1 mark)
13. Differentiate between AC and DC transmission (2 marks)
14. State the advantages and disadvantages of AC transmission (5 marks)
15. State the advantages and disadvantages of DC transmission? (5 marks)
16. State testing in line transmission. (2 marks)
17. State FOUR systems of power transmission. (4 marks)
18. State FOUR important components of an overhead transmission line. (4 marks)
19. Highlight FOUR advantages of pin-type insulator used in power transmission lines.
(4 marks)
20. (4 marks)
21. Explain TWO purposes of suspension insulators as used in high voltage power transmission.
(4 marks)
22. Outline THREE methods of improving string efficiency as used in design of transmission lines. (6 marks)
23. Explain TWO methods of reducing corona effect in an overhead transmission line.
(4 marks)
24. (4 marks)
25. Define the term “sag” as used in overhead lines. (2 marks)
26. Outline TWO factors which affect the sag of overhead transmission lines. (2 marks)
27. State THREE classifications of overhead transmission lines citing their voltage levels.
(6 marks)
28. Outline FOUR properties of conductor material used for transmission and distribution of electric power. (4 marks)
29. A 3-phase, 220 kV, 50 Hz transmission line consists of 1.5 cm radius conductor spaced 2 meters apart in equilateral triangular formation. If the temperature is 40°C and atmospheric pressure is 76cm, calculate the corona loss per km of the line. Take $m_0 = 0.85$. (10 marks)
30. A transmission line has a span of 275 m between level supports. The conductor has an effective diameter of 1.96 cm and weighs 0.865 kg/m. Its ultimate strength is 8060 kg. If the conductor has an ice coating of radial thickness 1.27 cm and is subjected to a wind pressure of 3.9 gm/cm^2 of projected area, calculate sag for a safety factor of 2. Weight of 1 c.c. of ice is 0.91 gm. (10 marks)